

Launching JupyterLab (Optional)

[!CAUTION]

If you are using the Virtual Machine (VM) we provided, JupyterLab is configured to launch automatically upon startup; you may simply skip this section.

- Access JupyterLab by entering the VM's IP address followed by the service port number (e.g., `IP:8888`).
- **JupyterLab Password: yahboom**

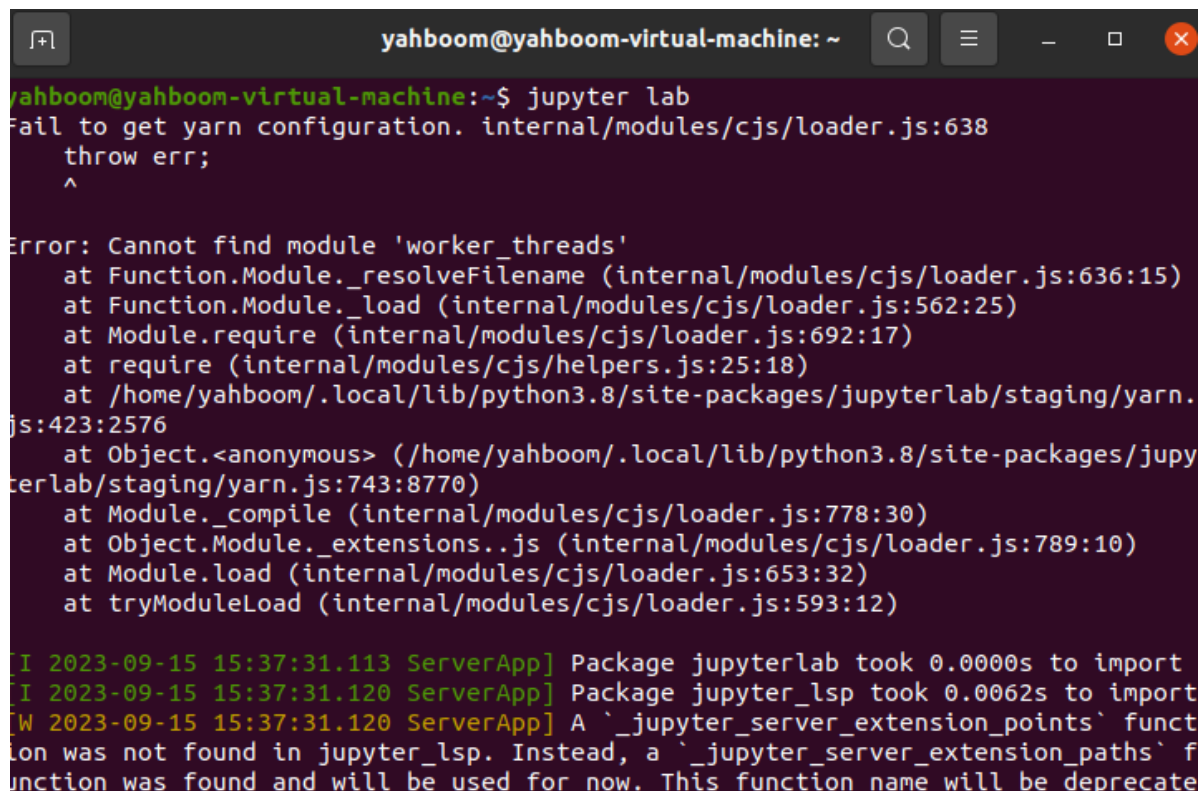
The following section outlines two methods for launching JupyterLab: the first is "Local Launch," which involves starting it directly within the VM's operating system; the second is "Remote Launch," which allows you to start and access JupyterLab remotely from another computer connected to the same local area network (LAN).

1. Local Launch

A prerequisite for opening JupyterLab locally is that the VM provided in the course materials must already be running on your computer. Open a terminal window and enter the following command:

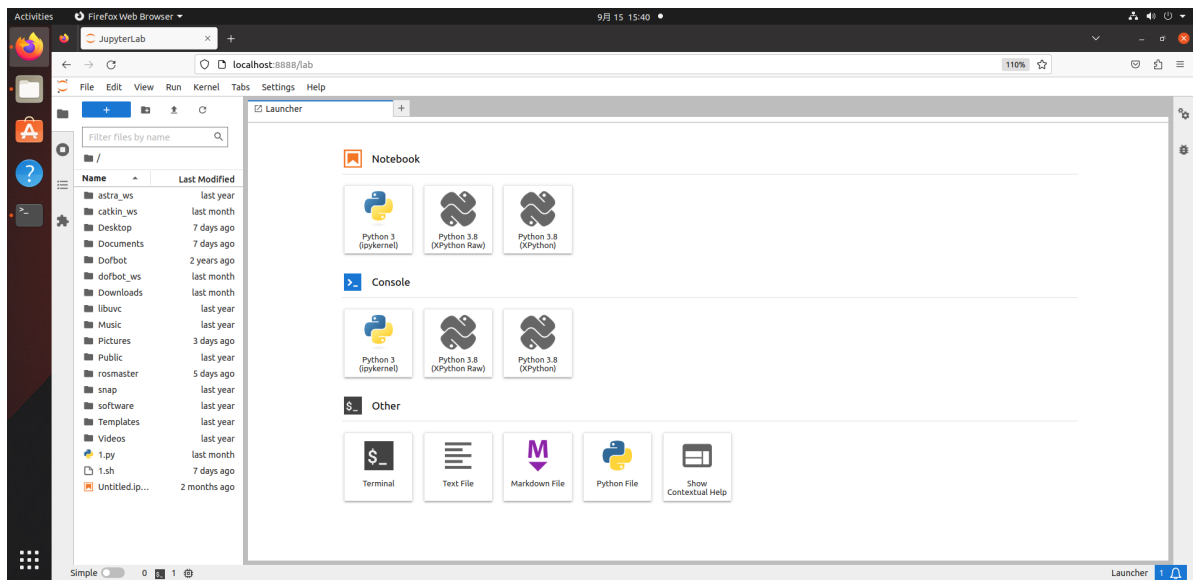
Note: The files displayed within the JupyterLab interface will correspond *only* to the files present in the specific directory where you opened the terminal.

```
jupyter lab
```



```
yahboom@yahboom-virtual-machine: ~  
yahboom@yahboom-virtual-machine:~$ jupyter lab  
Fail to get yarn configuration. internal/modules/cjs/loader.js:638  
  throw err;  
  ^  
  
Error: Cannot find module 'worker_threads'  
    at Function.Module._resolveFilename (internal/modules/cjs/loader.js:636:15)  
    at Function.Module._load (internal/modules/cjs/loader.js:562:25)  
    at Module.require (internal/modules/cjs/loader.js:692:17)  
    at require (internal/modules/cjs/helpers.js:25:18)  
    at /home/yahboom/.local/lib/python3.8/site-packages/jupyterlab/staging/yarn.  
js:423:2576  
    at Object.<anonymous> (/home/yahboom/.local/lib/python3.8/site-packages/jupy  
terlab/staging/yarn.js:743:8770)  
    at Module._compile (internal/modules/cjs/loader.js:778:30)  
    at Object.Module._extensions..js (internal/modules/cjs/loader.js:789:10)  
    at Module.load (internal/modules/cjs/loader.js:653:32)  
    at tryModuleLoad (internal/modules/cjs/loader.js:593:12)  
  
[I 2023-09-15 15:37:31.113 ServerApp] Package jupyterlab took 0.0000s to import  
[I 2023-09-15 15:37:31.120 ServerApp] Package jupyter_lsp took 0.0062s to import  
[W 2023-09-15 15:37:31.120 ServerApp] A `'_jupyter_server_extension_points'` func  
tion was not found in jupyter_lsp. Instead, a `'_jupyter_server_extension_paths'` f  
unction was found and will be used for now. This function name will be deprecate
```

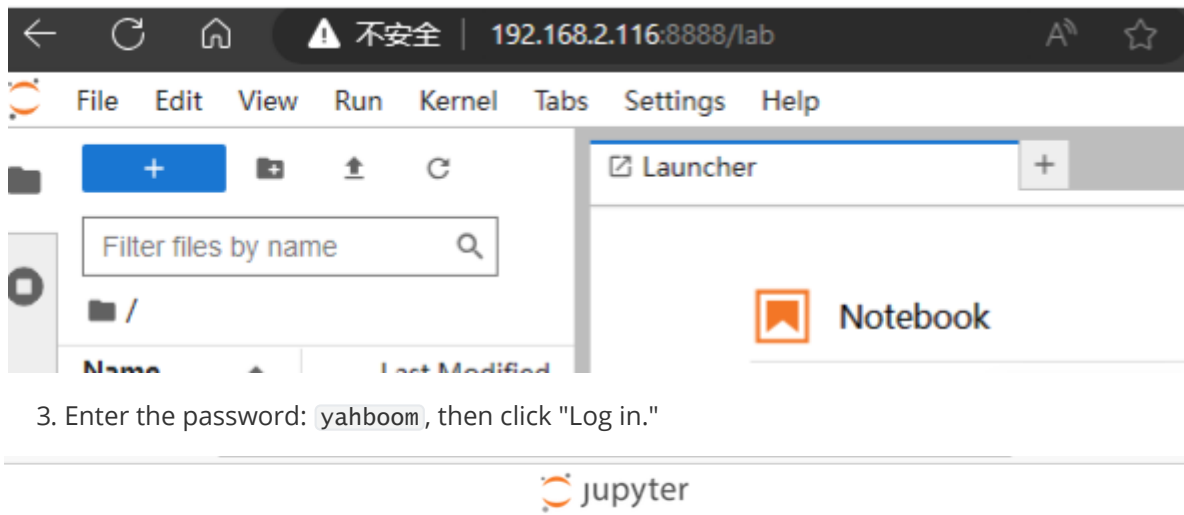
You may be prompted to enter a password upon your first login. The password is: `yahboom`. Subsequent logins should occur automatically. Upon successful login, the interface shown below will appear:



2. Remote Launch

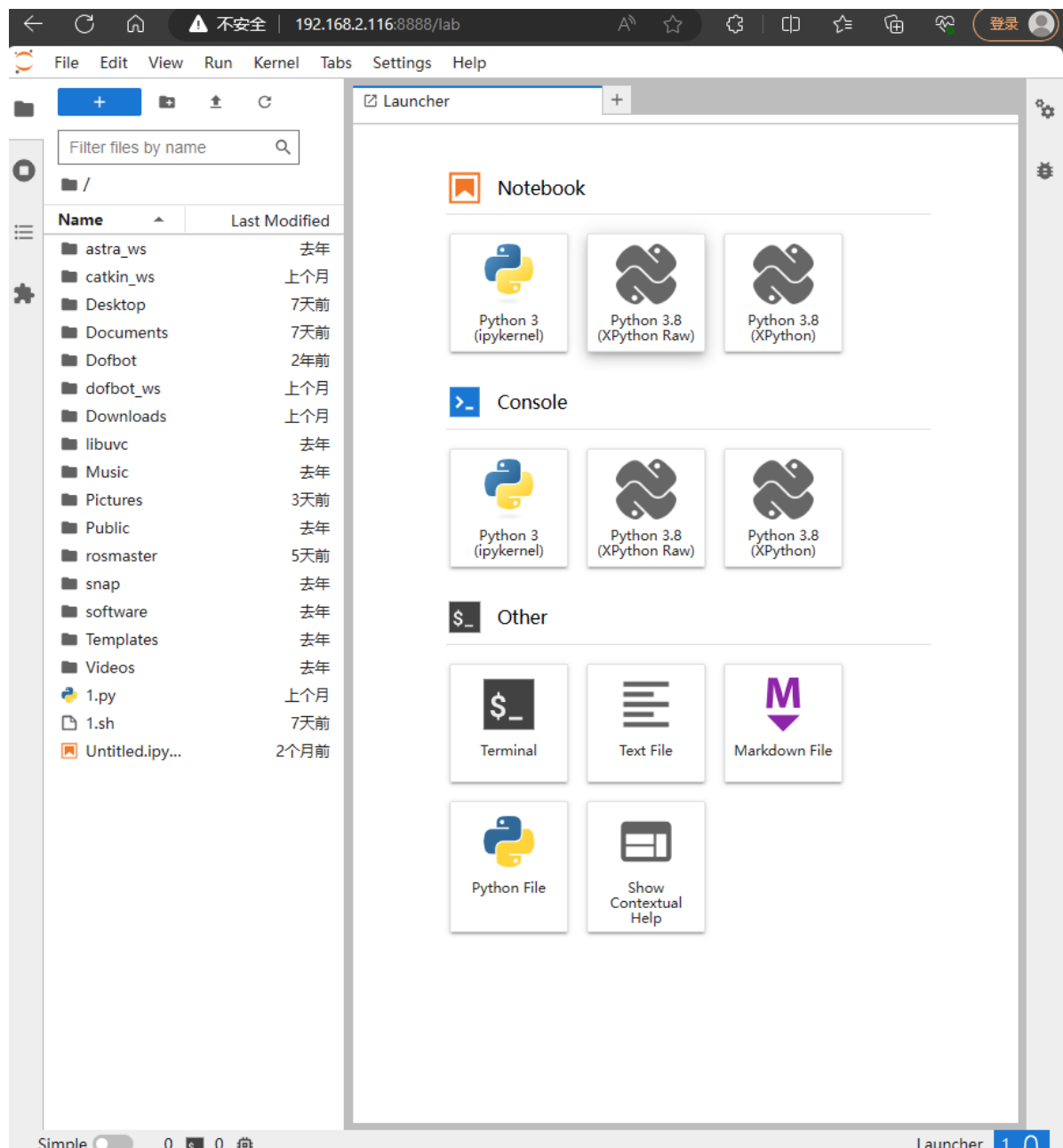
1. Determine the VM's current IP address. You can do this by opening a terminal window within the VM and entering the command `ifconfig`.
2. Open a web browser on your host computer (we recommend using Google Chrome). Enter the VM's IP address followed by the port number `8888`.

For example: `192.168.2.116:8888`



3. Enter the password: `yahboom`, then click "Log in."

Password: Log in



Note: The computer you are using for remote access must be connected to the *same local area network (LAN)* as the VM—that is, both devices must be connected to the same router.